

FROSTHOLD -- Feature & Mechanics Testing Checklist

Frostpunk x RimWorld Colony Survival Sim
Love2D 11.4 | Lua 5.1/LuaJIT | 1280x720
Generated: 2026-03-16
> This document lists every game feature and mechanic with its implementation status
> and testable acceptance criteria. Use the "Done" column to track QA passes.

1. CORE GAME LOOP

Game States (6)

Done	State	Test
	Start Menu	Menu loads, buttons respond, new game starts
	Planet Selection	7 planet cards display, arrow keys browse, enter confirms
	World Map (Hex)	Hex grid generates per-planet, landing zone selectable, info panel shows biome/resources/threat
	Colonist Drafting	Draft screen shows, colonists selectable, game begins
	Playing	Simulation runs, all systems ticking, input responsive
	Game Over	Triggers on all-dead, shows stats, restart/quit works

Time & Tick (5)

Done	Feature	Test
	20Hz fixed timestep	Simulation advances at consistent rate regardless of FPS
	Pause (Space)	All simulation stops, UI still responsive, unpause resumes
	Speed 1x (key 1)	Normal game speed
	Speed 2x (key 2)	Double speed, all systems scale correctly
	Speed 3x (key 3)	Triple speed, no desync or skipped events

Camera (6)

Done	Feature	Test
	WASD/arrow pan	Camera moves in all 4 directions
	Scroll wheel zoom	Zoom in/out with limits, centered on cursor
	Edge-of-screen pan	Camera moves when cursor touches window edges
	Window resize	Camera adjusts, no rendering artifacts, min 960x540
	Jump to entity	Click alert or colonist bar -> camera centers on target
	Minimap click	Click minimap -> camera jumps to that position

2. PLAYER CONTROLS & INPUT

Selection (6)

Done	Feature	Test
	Left-click select colonist	Single colonist selected, info panel opens
	Drag-box select	Green rectangle selects multiple colonists in area
	Shift+click add to selection	Adds/removes colonists from current selection
	N key cycle	Cycles through colonists alphabetically
	Click building	Building selected, inspection panel opens
	Click empty tile	Deselects current selection

Designation Tools (6)

Done	Tool	Key	Test
	Build Mode	B	Opens build menu, categories visible, buildings selectable
	Mine Designate	M	Drag marks tiles, blue overlay, colonists queue mining
	Stockpile Zone	Z	Drag paints zone, items get hauled to it
	Dumping Zone	X	Drag paints zone, waste hauled there
	Deconstruct	D	Drag marks buildings, colonists queue deconstruction
	Forage Designate	F	Drag marks tiles, colonists queue foraging

Context Menu (6)

Done	Target	Test
	Right-click colonist	Menu shows: medical, inspect options
	Right-click creature	Menu shows: hunt, force attack
	Right-click building	Menu shows: inspect, repair, deconstruct, set recipe
	Right-click item	Menu shows: haul, force eat, force haul
	Right-click rock/tree	Menu shows: force mine, force chop
	Right-click tile (drafted)	Selected colonist moves to tile

Hotkeys (16)

Done	Key	Action	Test
	ESC	Cancel/menu	Cancel build -> exit build mode -> pause menu
	TAB	Minimap toggle	Minimap appears/disappears at bottom-right
	F1	Tutorial toggle	Tutorial panel shows/hides
	F2	Thermal overlay	Heat map overlay visible on tiles
	F3	Pollution overlay	Pollution levels visible
	F4	Debug overlay	FPS, tick, entity counts, temperatures, power
	F5	Quick save	Save toast appears, file written
	F9	Quick load	Load toast appears, game state restored

	R	Research panel	Full-screen tech tree opens/closes
	P	Policy panel	Policy toggles open/close
	T	Trade panel	Merchant trade UI opens/closes
	E	Expedition view	Expedition map view opens/closes
	G	Farm panel	Farm management opens/closes
	V	Equipment panel	Gear assignment opens/closes
	H	Medical panel	Medical overview opens/closes
	C	Colony panel	Colony roster opens/closes
	L	Event log	Event history panel opens/closes
	Q	Quest panel	Quest tracker opens/closes

3. UI PANELS

Top Bar (5)

Done	Element	Test
	Resource display	Thermal Cores, Wood, Stone, Metal, Food, Water, Fuel with tooltips
	Research status	Current research name + progress bar
	Temperature display	Colony effective temp shown, updates with conditions
	Day/time display	Day counter + HH:MM advances correctly
	Speed buttons	1x/2x/3x + pause clickable, visual feedback on active

Colonist Bar (4)

Done	Element	Test
	Colonist cards	All living colonists shown with health/morale bars
	Status icons	Mental break (!), disease (+), injured (*), hungry (~), drafted (D), hypothermia (#)
	Card click	Selects colonist, opens info panel
	Hover tooltip	Shows name, HP, needs, status, current task

Alert System (15)

Done	Alert	Test
	FIRE	Appears on active fires, click jumps to fire
	RAID INCOMING	Appears when raid starts, click jumps to raid
	LOW FOOD	Triggers below food threshold
	LOW WATER	Triggers below water threshold
	LOW FUEL	Triggers below fuel threshold
	HYPOTHERMIA	Appears when colonist is hypothermic
	SUFFOCATION	Appears on low O2
	IDLE COLONISTS	Shows count of idle workers
	SICK	Appears on active disease
	MENTAL BREAK	Appears on breakdown
	REVOLT	Appears on high discontent
	POWER OUTAGE	Appears on brownout
	CRITICAL HEALTH	Appears on near-death colonist
	STARVATION	Appears on starving colonist
	HOPE CRITICAL	Appears on dangerously low hope

Colonist Info Panel -- 9 Tabs (9)

Done	Tab	Test
	Needs	5 need bars (warmth/food/water/rest/morale) with values, traits, backstory
	Health	Body parts list (HP/max), wounds, disease, hypothermia, addictions
	Social	Relationships list with opinion values, color-coded friend/rival
	Work	Priority matrix (10 jobs x 4 levels), skills list with levels
	Schedule	24hr blocks (work/eat/sleep/free), clickable to change
	Gear	Weapon/armor/accessory slots, equip/unequip
	Bio	Backstory, traits, personality
	Combat	Combat skills, history
	Log	Personal event history

Build Menu (5)

Done	Feature	Test
	10 build categories	Structure/Heating/Power/Lighting/Air/Production/Colony/Logistics/Logic/Defense
	Building cards	Show name, cost breakdown, power draw
	Affordability color	Green = affordable, red = too expensive
	Click to select	Ghost overlay appears on map at cursor
	R to rotate	Direction cycles right->down->left->up

Building Inspection (6)

Done	Feature	Test
	Name + description	Building name and desc shown on selection
	Durability bar	HP bar with numeric value
	Machine status	Recipe, progress %, power status
	Reactor status	Power level, fuel %, overdrive, stress %
	Steam hub status	Active/inactive indicator
	Battery charge	Charge % with bar

Full-Screen Panels (8)

Done	Panel	Key	Test
	Research	R	5-tier tree, click to select/begin, cancel button, progress bar
	Policies	P	Toggle laws on/off, effects summary (work speed, food drain, morale)
	Trade	T	Buy/sell items, thermal cores currency, merchant info + timer
	Farm	G	Farm plots, growth status, crop picker, planting
	Equipment	V	Colonist gear assignment, 3 slots, equip/unequip
	Medical	H	Wound list, disease, surgery queue, body part HP
	Colony	C	Full roster, sortable columns, click to select/center
	Expedition	E	Off-map tilemap, fog of war, POI markers, progress

Pause Menu (4)

Done	Feature	Test
	ESC opens menu	Shows PAUSED, colony info, day count
	Resume button	Unpauses game
	Save/Load buttons	F5/F9 equivalent from menu
	Quit button	Returns to main menu or exits

4. PLANET SYSTEM

Planet Selection UI (7)

Done	Feature	Test
	7 planet cards render	All 7 planets shown as horizontal cards with color, name, subtitle, difficulty badge
	Arrow key browsing	Left/right arrows cycle through unlocked planets
	Mouse click selection	Click planet card selects it
	Planet descriptions	Extended description shown for selected planet at bottom
	Difficulty badges	Standard/Easy/Hard/Very Hard/Extreme/Narrative labels correctly colored
	Locked planet handling	Locked planets show "Coming Soon" overlay, not selectable
	Configure Deployment button	Confirm advances to world_map phase with correct planet

Planet Definitions (7)

Done	Planet	Difficulty	Test
	Erebus (Frozen World)	Standard	All nil overrides, falls through to defaults, 6 scenarios
	Rhea-2 (Scorched Desert)	Hard	Heat/water management, twin suns, 3 scenarios
	Morvos (Acid World)	Very Hard	Corrosion, toxic atmosphere, 2 scenarios
	Nerthus-9 (Ocean World)	Hard	Pressure, flooding, underwater, 2 scenarios
	Paxtera Prime (Temperate)	Easy	Mild seasons, standard colony sim, 5 scenarios
	Nemaea (Dead World)	Extreme	Vacuum, radiation, automatons, 2 scenarios
	Gaia A^1x (Fall of Fortuna)	Narrative	Baldrungen escalation, fixed ending, 1 scenario

World Map / Hex Grid (8)

Done	Feature	Test
	Hex grid generation	Radius-3 grid (37 hexes) generated with planet-specific biomes
	Biome assignment	Weighted noise-based biome picking from planet palette
	Resource distribution	Sparse/normal/rich assigned per hex from noise + biome bias
	Threat levels	Low/medium/high/extreme assigned per hex, edges higher
	Temperature modifier	Biome properties set temp modifier per hex
	Elevation	0-1 elevation from noise displayed per hex
	Info panel	Right-side panel shows biome name, temp, elevation, resources, threat
	Confirm Landing Zone	Selected hex stores landingZone on GameState, advances to drafting

Per-Planet Biomes (35)

Done	Planet	Biomes	Test
	Erebus	Tundra, Glacier, Volcanic, Frozen Marsh, Frozen Forest	5 biomes with correct colors and weights
	Rhea-2	Dune Sea, Canyon, Oasis, Badlands, Salt Flat	5 biomes, desert palette
	Morvos	Acid Basin, Fungal Grove, Rock Platform, Toxic Marsh, Spore Field	5 biomes, acid palette
	Nerthus-9	Volcanic Island, Coral Reef, Active Volcano, Deep Ocean, Atoll	5 biomes, ocean palette
	Paxtera Prime	Grassland, Forest, Farmland, Hills, Wetland	5 biomes, temperate palette
	Nemaea	Crater Field, Dyson Ruins, Irradiated Wastes, Ship Wreckage, Regolith Plains	5 biomes, dead world palette
	Gaia A^1x	Meadow, Dense Forest, Corrupted Zone, River Valley, Highlands	5 biomes, lush+corrupt palette

Per-Planet Seasons (28)

Done	Planet	Seasons	Test
	Erebus	Deep Winter / Late Winter / Thaw / Freeze	Default 15-day seasons, existing behavior
	Rhea-2	Scorching (55C) / Dry (35C) / Dust Storm (40C) / Cool (20C)	15-day desert seasons, starts in Dry
	Morvos	Acid Rain / Toxic Calm / Spore Bloom / Corrosion Peak	20-day acid seasons, starts in Toxic Calm
	Nerthus-9	Monsoon / Low Tide / Storm Season / Calm Waters	18-day ocean seasons, starts in Low Tide
	Paxtera Prime	Spring (12C) / Summer (25C) / Autumn (10C) / Winter (-5C)	15-day temperate seasons, starts in Spring
	Nemaea	Solar Maximum (120C) / Eclipse (-80C) / Solar Minimum (5C) / Debris Season (-20C)	30-day extreme seasons, starts...
	Gaia A^1x	Spring (15C) / Summer (28C) / Autumn (8C) / Corruption (-5C)	15-day seasons with narrative escalation, starts...

Per-Planet Atmosphere (7)

Done	Planet	O2/C02	Test
	Erebus	100/0 (breathable)	Default atmosphere
	Rhea-2	100/0 (breathable)	Desert but breathable

	Morvos	70/15 (toxic)	Reduced O2, CO2 buildup, sealed rooms needed
	Nerthus-9	100/0 (breathable)	Surface breathable, underwater needs O2
	Paxtera Prime	100/0 (breathable)	Standard atmosphere
	Nemaesa	0/0 (vacuum)	No atmosphere, sealed habs + suits mandatory
	Space	0/0 (vacuum)	Ship-only atmosphere

Per-Planet Secrets (7)

Done	Planet	Secret Types	Test
	Erebus	Frozen colonist, thing mimic, precursor artifact, sealed cache, dormant nest	Default secrets
	Rhea-2	Buried Cryopod, Sandstone Cache, Sun Shrine, Sand Wurm Nest	4 desert secrets
	Morvos	Sealed Pod, Toxic Cache, Spore Node, Corrosion Nest	4 acid secrets
	Nerthus-9	Sunken Pod, Coral Cache, Depth Beacon, Kraken Nest, BioVault Vessel, Mammona Rig, Sunken Lab, Cargo Boat, Trawl...	
	Paxtera Prime	Abandoned Shelter, Supply Drop, Old Bunker, Wildlife Den	4 temperate secrets
	Nemaesa	Automaton Pod, Dyson Cache, Signal Beacon, Drone Nest	4 dead world secrets
	Gaia A^1x	Forest Shelter, Growth Cache, Corruption Node, Husk Nest	4 Baldrungen secrets

Per-Planet Tuning (7)

Done	Planet	Raid/Weather Tuning	Test
	Erebus	Default tuning	Baseline difficulty
	Rhea-2	Budget base 25, +2.5/day; harshness 0.6-2.2	Hot raids, harsh weather
	Morvos	Budget base 20, +1.5/day; harshness 0.8-2.5	Lower raid frequency, extreme weather
	Nerthus-9	Budget base 25, +2/day; harshness 0.5-2.0	Standard raids with flooding
	Paxtera Prime	Budget base 25, +2/day; harshness 0.3-1.5	Easy weather
	Nemaesa	Budget base 40, +3/day; harshness 1.0-3.0; human_chance=0	Automaton-only raids, extreme
	Gaia A^1x	Budget base 20, +4/day; swarm_chance 2x; harshness 0.3-2.0	Escalating insectoid swarms

5. PER-PLANET MECHANICS

Rhea-2 Heat Management (4)

Done	Feature	Test
	Daytime surface temp 55C (scorching)	Colonists take heat damage outdoors during scorching season
	Underground base temp 25C	Underground provides shelter from heat
	Night operations	Daylight hours extended (5-21 in scorching), night is safer
	Water scarcity	Oasis tiles as water sources, condensation traps functional

Morvos Corrosion System (6)

Done	Feature	Test
	Planet check	Corrosion system activates only when GameState.planet == 'morvos'
	Outdoor corrosion	Buildings outdoors lose 0.1 durability/sec
	Indoor unsealed corrosion	Indoor unsealed buildings lose 0.03 durability/sec
	Sealed room immunity	Buildings in sealed rooms take no corrosion damage
	Weather multiplier	Acid storm and corrosive fog double corrosion rate (2x)
	Season multiplier	Corrosion peak season doubles corrosion regardless of weather
	Building destruction	Buildings at 0 durability destroyed and entity removed

Nerthus-9 Underwater Pressure (8)

Done	Feature	Test
	Planet check	Pressure system activates only on nerthus_9
	Depth-based pressure	1.0 pressure per depth level, linear scaling
	Surface safe	Depth 0 or above has zero pressure
	Sealed room protection	Enclosed rooms with no flooding negate pressure
	Suit resistance	Diving suit pressure rating reduces effective pressure (0-100%)
	Pressure damage	0.15 HP/sec per unit of unresisted pressure
	Crush threshold	Pressure above 4.0 causes rapid death (2.0 HP/sec per unit)
	Cause of death	Crush deaths log as 'pressure_crush'

Nerthus-9 Flooding (4)

Done	Feature	Test
	Flow rate 2.0	Water flows between tiles at 2x normal rate on Nerthus-9
	Evaporation rate 0.3	Slow evap in ocean environment
	Monsoon floods	Monsoon season weather bias heavily favors heavy_rain + storms with flooding
	Hurricane flooding	Hurricane weather type sets floods=true, injects water into outdoor tiles

Nemaesa Vacuum & Radiation (6)

Done	Feature	Test
	No atmosphere	O2 = 0, CO2 = 0; colonists suffocate without sealed habs
	Ambient radiation	0.02 dose/tick; lethal dose 5.0
	Temperature extremes	Solar Maximum 120C, Eclipse -80C
	No organic wildlife	All creatures are mechanical automatons
	No hermes system	hermes.enabled = false
	Dyson Sphere themes	Solar flare, meteor shower, radiation burst weather unique to Nemaesa

Gaia A^1x Baldrungen Escalation (10)

Done	Feature	Test
	Planet check	Baldrungen system activates only when GameState.planet == 'gaia_a1x'

	Corruption season detection	Season transition to 'corruption' increments corruption cycles
	Cycle 1 waves	2-4 husk_crawlers spawn from map edges
	Cycle 2 waves	3-5 mixed husk_crawlers + bone_beetles
	Cycle 3 waves	4-7 mixed crawlers + beetles + rot_wasps
	Cycle 4+ waves	6-10+ massive swarms with brood_mothers and the_emergence
	Spawn timer	30-60s random interval between wave spawns during corruption
	Artifact shielding	3 contained artifacts create ward, cap spawn intensity at cycle 3
	Final cycle (5)	Baldrungen fully awakens; if shielded, colony survives; if not, game enters 'baldrungen_fall' mode
	Alerts	Corruption Wave/Baldrungen Swarm/The Emergence alerts fire at appropriate severity

Paxtera Prime Temperate Colony (3)

Done	Feature	Test
	Mild seasons	Spring 12C to Winter -5C; nothing like Erebus extremes
	Human-centric raids	Standard raid system with all factions active
	Full scenario selection	5 scenarios available (crashlanded, lone_wanderer, lost_tribe, rich_explorer, naked_brutality)

Space (Non-Starting Planet) (3)

Done	Feature	Test
	Not selectable as starting planet	scenarios = {} prevents selection
	Vacuum environment	O2 = 0, outdoor temp snap 1.0, underground base temp -270C
	Ambient radiation	0.01 dose, lethal 5.0

6. COLONIST SYSTEMS

Needs (5)

Done	Need	Test
	Warmth	Decays when cold, satisfied by heated rooms, hypothermia stages trigger
	Food	Decays over time, colonist seeks food, starvation -> death
	Water	Decays over time, colonist seeks water
	Rest	Decays over time, colonist seeks bed, exhaustion effects
	Morale	Affected by environment, social, events, triggers mental breaks at low

Hypothermia (5 stages)

Done	Stage	Threshold	Test
	Normal	>= 60 warmth	No effects
	Chilled	>= 40	Minor speed penalty
	Cold	>= 20	Moderate penalty, alert shows
	Hypothermic	>= 10	Severe penalty, health damage
	Severe	< 10	Near-lethal, rapid health loss

Work Priority System (10 job types)

Done	Job	Test
	Mining	Colonist mines designated tiles, yields resources
	Building	Colonist constructs placed blueprints
	Hauling	Colonist moves items to stockpiles
	Operating	Colonist operates production machines
	Cooking	Colonist uses kitchen to make meals
	Hunting	Colonist targets designated creatures
	Research	Colonist works at research bench, progress advances
	Medical	Colonist treats wounded/sick colonists
	Cleaning	Colonist cleans filthy areas
	Firefighting	Colonist auto-extinguishes active fires

Scheduling (4)

Done	Block	Test
	Work	Colonist performs assigned job priorities
	Eat	Colonist seeks food during eat blocks
	Sleep	Colonist seeks assigned bed
	Free	Colonist satisfies needs or idles

Skills & XP (10)

Done	Skill	Test
	Mining	Levels up from mining, affects speed
	Building	Levels up from construction, affects speed/quality
	Cooking	Levels up from cooking, affects food quality
	Medical	Levels up from treatment, affects success rate
	Combat	Levels up from fighting, affects hit chance/damage
	Research	Levels up from bench work, affects research speed
	Farming	Levels up from farm work, affects growth/yield
	Crafting	Levels up from crafting, affects quality
	Passion system	Interested (1.5x XP) / Passionate (2x XP) visible and working
	Skill rust	Unused skills get -20% effectiveness, clears after 3 uses

Traits & Backstories (3)

Done	Feature	Test

	Trait generation	Colonists spawn with random traits from adlib.lua
	Trait effects	Traits modify needs/work/combat/social as defined
	Backstory generation	Colonists get procedural backstories from adlib system

Scar Traits (10)

Done	Scar	Trigger	Test
	frostblood	Survive hypothermia	+cold resistance permanent
	battle_hardened	Survive raid at low HP	+combat mod permanent
	iron_stomach_scar	Survive starvation	+food efficiency permanent
	plague_immune	Survive disease	+disease immunity permanent
	mental_resilience	Survive mental break	+morale stability permanent
	efficient_metabolism	Survive long hunger	+food mod permanent
	deep_scars	Major wound recovery	Stat modification permanent
	pyromaniac_reformed	Survive pyro break	Pyro trait removed permanent
	blizzard_walker	Survive whiteout	+speed in snow permanent
	Other scars	Various hardships	Verify each trigger fires correctly

Draft Mode (3)

Done	Feature	Test
	Draft toggle	Selected colonist enters manual control
	Move command	Right-click tile -> colonist moves there
	Attack command	Right-click creature -> colonist attacks

7. SOCIAL & MORALE

Social System (5)

Done	Feature	Test
	Opinion matrix	Symmetric opinions between colonist pairs (-100 to +100)
	Proximity gain	Opinion increases when colonists are near each other
	Friends (>50)	Morale buff when friend is nearby
	Rivals (<-30)	Morale penalty, fight chance when adjacent
	Grief	-20 morale for 2 days when friend dies

Hope & Discontent (6)

Done	Feature	Test
	Hope meter (0-100)	Visible in UI, drifts toward 50
	Discontent meter (0-100)	Visible in UI, drifts toward 20
	Death impact	Colonist death drops hope by ~15
	Good events raise hope	Good meal, building, etc. raise hope
	Despair tracking	Extended low hope -> colony-wide mental breakdown
	Revolt tracking	Extended high discontent -> colonist rebellion

Mental Breaks (8)

Done	Break	Test
	Pyromaniac	Low morale -> colonist starts fires
	Coward	Stress -> colonist flees, hides
	Glutton	Stress -> colonist binge eats food stores
	Berserk	Stress -> colonist attacks others
	Ruin drawn	Stress -> colonist wanders toward map edge
	Signal listener	Stress -> colonist fixates, stops working
	Quarantine panic	Stress -> colonist panics, runs
	Break recovery	Breaks have duration, colonist returns to normal

Addiction & Drugs (8)

Done	Drug	Test
	Spike	Drug consumed, effects applied, addiction risk
	Stardust	Drug consumed, effects applied, addiction risk
	Drift	Drug consumed, effects applied, addiction risk
	Smog	Drug consumed, effects applied, addiction risk
	Shards	Drug consumed, effects applied, addiction risk
	Glimpse	Drug consumed, effects applied, addiction risk
	Surge	Drug consumed, effects applied, addiction risk
	Voidbloom	Drug consumed, effects applied, addiction risk
	Withdrawal	Addicted colonist without drug suffers penalties

Policies (6)

Done	Policy	Test
	Extended Shifts	Toggle on -> work speed up, morale down
	Rationing	Toggle on -> food consumption down, morale down
	Martial Law	Toggle on -> discontent suppressed, hope penalty
	Emergency Protocol	Toggle on -> emergency effects active
	Quota Compliance	Toggle on -> production bonus, morale cost
	Blackout Protocol	Toggle on -> heat signature reduced, warmth penalty

8. BUILDING & CONSTRUCTION

Placement (6)

Done Feature Test
----- ----- -----
Ghost preview Semi-transparent building follows cursor
Cost check Red ghost if resources insufficient
Collision check Cannot place on occupied tiles
Multi-tile buildings Large buildings (reactor 3x3) place correctly
Line placement Pipes/belts/walls support drag-line placement
Build task created Colonist queues construction after placement

Building Categories -- Verify All Buildable (10)

Done Category Count Test
----- ----- ----- -----
Structure 11 Walls, floors, doors, columns all placeable and functional
Heating 3 Campfire, heater, steam hub produce heat
Power 35+ All generators produce power when fueled/conditions met
Lighting 3 Torch, lamp, sun lamp illuminate areas
Ventilation 4 Air intake/exhaust/purifier/scrubber affect room atmosphere
Production 14 All machines accept recipes and produce items
Colony 2+ Cloning vat, radio beacon, beds, memorial functional
Logistics 13+ Conveyors, inserters, pipes, tanks placeable and transport
Defense 25+ Turrets, traps, fortifications, laser fences placeable
Endgame 4 Victory buildings constructible

Production Machines (14)

Done Machine Test
----- ----- -----
Sawmill Raw wood -> lumber
Smelter Raw ore -> metal ingots
Forge Metal -> steel, components, tools
Kitchen Raw food -> cooked meals
Workbench General crafting recipes
Loom Fiber/hide -> cloth
Tannery Raw hides -> leather
Smokehouse Meat -> jerky
Drug Lab Herbs -> drugs/medicine
Butcher Table Corpses -> meat + hide
Refinery Raw materials -> fuel
Oil Refinery Crude oil -> fuel
Med Bench Herbs -> medicine/bandages
Vehicle Workbench (CUT) Archived stub. Vehicles are not in shipped scope.

Recipe Selection (2)

Done Feature Test
----- ----- -----
Right-click machine -> set recipe Recipe list appears, selection applies
Machine produces selected recipe Inputs consumed, outputs created, progress visible

Recreation Buildings (6)

Done Building Test
----- ----- -----
Bonfire Outdoor gathering, 6 capacity, heat+light output, 0.08 joy/tick
Card Table 2 capacity, social bonds, 0.12 joy/tick
Tavern Bar 4 capacity, consumes food, 0.15 joy/tick
Sparring Ring 2 capacity, trains combat, 0.10 joy/tick
Library 3 capacity, boosts research, 0.06 joy/tick
Radio Set 4 capacity, plays music, 0.10 joy/tick, requires 5W power

Storage Buildings (7)

Done Building Test
----- ----- -----
Crate Basic wood storage, no protection
Locker Metal, insulated storage
Shelf Wide wall-mounted, open storage
Chest Sealed metal, weather-protected
Cold Storage Unit Powered refrigeration, 20W draw
Lead-lined Vault Radiation-shielded hazmat storage
Bulk Silo 3x3 enormous capacity for single category

Misc Buildings (7)

Done Building Test
----- ----- -----
Medical Bench Crafts medicine and medical supplies
Deep Drill Automated deep extraction, 50W, attracts creatures
Expedition Table Plan and launch off-map expeditions
Quest Board Post and track colony objectives
Cryo Pod Suspends colonist indefinitely, 25W draw
Scrubber Reduces pollution in nearby tiles, 20W draw
Sump Pump Drains water from flooded rooms, 15W draw

Containment Buildings (2)

Done Building Test

|-----|-----|-----|
| | Containment Cell | Powered quarantine for live subjects, 22W draw |
| | Anomaly Locker | Cold storage for artifacts and sealed anomaly samples, 12W draw |

Structural Support (3)

Done	Building	Test
	Wood Column	Cheap support, 8-tile span, degrades over time
	Stone Column	Durable, 14-tile span, no degradation
	Reinforced Column	Maximum support, 26-tile span

9. UNDERWATER & OCEAN BUILDINGS (Nerthus-9)

Pressure Infrastructure (7)

Done	Building	Test
	Airlock	Sealed transition chamber, cycles open/closed, 10W draw
	Pressure Dome	3x3 sealed dome, pushes water out of 5x5 area, 40W draw
	Diving Bell	2x2 descent chamber, refills suit O2, 25W draw
	Submersible	Sealed pod, carries 2 between depths, pressure-rated, 35W draw
	Electrolyzer (O2 Gen)	Splits water into breathable oxygen, 30W draw
	Bilge Pump	High-capacity water extraction, 25W draw
	Condensation Trap	Passive clean water collection, no power

Underwater Habitats (5)

Done	Building	Test
	Underwater Habitat	2x2 sealed quarters, insulated, 15W draw
	Hab Pod	1x1 compact sealed pod for one colonist, 8W draw
	Hab Block	3x3 large sealed habitat module, 25W draw
	Sealed Wall	Pressure-rated wall, blocks water/gas
	Glass Dome	Transparent pressure dome, morale bonus, blocks water/gas

Floating Surface Structures (4)

Done	Building	Test
	Pontoon Platform	Floating wood platform, extends buildable area over water
	Reinforced Dock	Metal-braced floating platform, supports heavier buildings
	Fishing Platform	2x1 floating nets, generates food passively from water tiles
	Tidal Generator	2x2 ocean current power, 70W output

Underwater Production (4)

Done	Building	Test
	Kelp Farm	2x1 underwater hydroponics, grows kelp for food/biomass, 10W draw
	Desalination Unit	Converts seawater to fresh water, 20W draw
	Seabed Drill	2x2 underwater mining rig, 60W draw, attracts predators, underwaterOnly
	Thermal Vent Tap	2x2 taps volcanic vent, 110W output + heat

Underwater Power (2)

Done	Building	Test
	Thermal Vent Tap	Taps ocean floor volcanic vent, 110W, some heat
	Current Turbine	Underwater turbine, 45W constant output from currents

10. WEATHER SYSTEM

Erebus Weather (7 types)

Done	Weather	Test
	Clear	Normal conditions, full visibility
	Overcast	-5C temp, 0.85 visibility, light precip
	Snowfall	-10C temp, 0.6 visibility, snow particles
	Blizzard	-25C temp, 0.2 visibility, lightning
	Whiteout	-35C temp, 0.05 visibility, lethal
	Warm Front	+15C temp, rain, golden tint
	Aurora	-5C, 0.95 visibility, morale buff, shimmer effect

Rhea-2 Desert Weather (3 types)

Done	Weather	Test
	Sandstorm	+10C, 0.1 visibility, 0.9 solar penalty, orange-brown tint
	Heat Wave	+20C, 0.85 visibility, no solar penalty
	Dust Devil	+5C, 0.6 visibility, 0.2 solar penalty

Morvos Acid Weather (4 types)

Done	Weather	Test
	Acid Storm	-5C, 0.15 visibility, 0.85 solar penalty, 2x corrosion
	Corrosive Fog	0C, 0.25 visibility, 1.5x corrosion
	Toxic Haze	+3C, 0.5 visibility, green tint
	Spore Cloud	+2C, 0.4 visibility, yellow-green tint

Nerthus-9 Ocean Weather (3 types)

```
| Done | Weather | Test |
|-----|-----|-----|
| | Hurricane | -8C, 0.08 visibility, floods, 3x wind, lightning |
| | Heavy Rain | -3C, 0.35 visibility, floods |
| | Storm | -5C, 0.3 visibility, 2x wind, lightning |
```

Paxtera/Gaia Temperate Weather (2 types)

```
| Done | Weather | Test |
|-----|-----|-----|
| | Rain | -2C, 0.7 visibility, light precipitation |
| | Spore Fall (Gaia only) | -3C, 0.35 visibility, corruption-themed |
```

Nemaea Dead World Weather (3 types)

```
| Done | Weather | Test |
|-----|-----|-----|
| | Solar Flare | +40C, 0.7 visibility, bright orange overlay |
| | Meteor Shower | -10C, 0.5 visibility, structural damage |
| | Radiation Burst | +5C, 0.8 visibility, pale green glow, radiation dose |
```

Weather Mechanics (6)

```
| Done | Feature | Test |
|-----|-----|-----|
| | Markov transitions | Weather changes via weighted transition table |
| | Seasonal bias | Season affects weather probability weights |
| | Planet-specific merging | Planet weather types merge into base type table at init |
| | Harshness scaling | Severe weather duration scales with harshness, calm weather inversely |
| | Weather particles | Type-specific colored particles (snow=white, sand=orange, acid=green, rain=blue) |
| | Weather overlays | Type-specific screen-space tints render correctly |
```

Blood Rain (special)

```
| Done | Feature | Test |
|-----|-----|-----|
| | Blood Rain weather type | +5C temp, 0.55 visibility, crimson tint, heavy flooding |
| | Easter egg trigger | First rain event becomes blood rain if Fischbach colonist exists |
| | Blood rain particles | Vertical red streak particles render |
```

11. THERMAL & ENVIRONMENT

Thermal Diffusion (5)

```
| Done | Feature | Test |
|-----|-----|-----|
| | Room temperature | Enclosed rooms have distinct temperatures |
| | Heat sources | Campfire/heater/reactor raise room temp |
| | Wall insulation | Insulated walls retain heat better than wood/stone |
| | Wind chill | Outdoor tiles affected by weather wind |
| | Thermal overlay (F2) | Heat map visible, hot=red, cold=blue |
```

Generator / Reactor (6)

```
| Done | Feature | Test |
|-----|-----|-----|
| | Placement | 3x3 reactor places correctly |
| | 4 power levels | Each level increases heat + fuel consumption |
| | Fuel consumption | Fuel depletes, reactor shuts off when empty |
| | Overdrive | Activates, stress climbs, increased output |
| | Meltdown | Stress reaches 100% -> explosion, damage radius, hope penalty |
| | Steam hubs | Extend reactor heat to distant tiles |
```

Atmosphere (5)

```
| Done | Feature | Test |
|-----|-----|-----|
| | Per-room O2/CO2 | Rooms track gas levels independently |
| | CO2 buildup | Sealed rooms with fire/colonists accumulate CO2 |
| | Suffocation | Low O2 damages colonists, alert triggers |
| | Air intake | Pulls outside air into room |
| | Air purifier | Filters CO2, requires power |
```

Lighting (3)

```
| Done | Feature | Test |
|-----|-----|-----|
| | Dark rooms | Rooms without light sources have darkness penalty |
| | Light sources | Torch/lamp illuminate radius, penalty removed |
| | Sun lamp | Enables crop growth indoors |
```

Seasons -- Erebus Default (4)

```
| Done | Season | Temp | Test |
|-----|-----|-----|-----|
| | Deep Winter | -60C | 15 days, lowest temps, reduced raids |
| | Late Winter | -45C | 15 days |
| | Thaw | -15C | 15 days, meltwater floods possible, forage bonus |
| | Freeze | -30C | 15 days, temps dropping again |
```

12. POWER SYSTEM

Power Grid (6)

Done	Feature	Test
-----	-----	-----
	Union-find connectivity	Buildings on same grid share power
	Consumer priority	Critical/normal/low priority shedding during brownout
	Brownout	Insufficient power -> low-priority buildings shut off
	Power conduits	Connect distant buildings to same grid
	Power switch	Toggle isolates grid sections
	Battery charge/discharge	Batteries store excess, release during deficit

Power Generators (35)

Done	Generator	Output	Test
-----	-----	-----	-----
	Campfire	10W	Burns raw_wood
	Coal Burner	30W	Burns coal
	Thermal Generator	80W	Burns thermal_core
	Fuel Cell Generator	50W	Burns fuel_cell
	Lava Vent Tap	100W	No fuel, placed on vent
	Solar Panel	40W	Intermittent solar
	Tracking Solar Array	65W	Intermittent solar
	Concentrated Solar	100W	Intermittent solar
	Wind Turbine	55W	Intermittent wind
	Large Wind Turbine	90W	Intermittent wind
	Advanced Turbine	130W	Intermittent wind
	Geothermal Vent	120W	No fuel
	Gas Burner	45W	Burns coal, high CO2
	Nuclear Reactor	250W	Burns thermal_core, meltdown risk
	Chemical Burner	60W	Burns components
	Bio Reactor	35W	Burns food
	Mini Reactor	90W	Burns thermal_core, minor meltdown risk
	Steam Turbine	65W	Intermittent heat
	Hand Crank	12W	Manned, no fuel
	Treadmill	25W	Manned, forced labor
	Chain Gang Wheel	50W	Manned, forced labor, 3 crew
	Waste Incinerator	40W	Burns corpses, high CO2+heat
	Hydrogen Fuel Cell	70W	Intermittent water
	Lightning Rod	200W	Intermittent storm
	Thermopile	25W	Intermittent thermal diff
	Ichor Burner	55W	Burns eldritch_ichor
	Cryo-Kinetic Engine	60W	Intermittent cold
	Methane Digester	40W	Burns food, produces heat
	Plasma Arc Reactor	180W	Burns steel, meltdown risk, massive heat
	Stirling Engine	45W	Intermittent thermal diff
	Penrose Engine	100W	Endgame exotic, no fuel
	Dynamo	20W	Manned, any worker
	Peat Burner	25W	Burns any, produces heat
	Fusion Reactor	400W	Endgame, no fuel, minor meltdown risk
	Ichor Converter	40W	Byproduct processor
	Fuel Generator	60W	Burns refined fuel
	Tidal Generator	70W	Intermittent water, ocean surface
	Thermal Vent Tap	110W	No fuel, ocean floor
	Current Turbine	45W	Intermittent water, underwater

13. COMBAT & DEFENSE

Melee Combat (4)

Done	Feature	Test
-----	-----	-----
	Melee attack	Colonist/creature deals damage on adjacent tile
	Weapon damage	Equipped weapon modifies damage output
	Body part targeting	Hits land on specific body parts
	Armor absorption	Equipped armor reduces incoming damage

Ranged Combat (5)

Done	Feature	Test
-----	-----	-----
	Projectile firing	Colonist/turret fires projectile at target
	Hit probability	Affected by skill, distance, cover
	Cover system	Fortifications reduce hit chance on defenders
	Weapon types	Bow/crossbow/revolver/rifle each have distinct stats
	Ammo consumption	Ranged weapons consume correct ammo type

Body & Wounds (5)

Done	Feature	Test
-----	-----	-----
	6 body parts	Head, torso, L/R arm, L/R leg tracked independently
	Wound types	Cuts, burns, frostbite, fractures applied correctly
	Wound treatment	Medical colonist treats wounds, reduces infection
	Bleeding	Untreated wounds cause ongoing health loss
	Part destruction	Body part at 0 HP -> disability/death

Turrets -- Spot Check (5)

```
| Done | Turret | Test |
|-----|-----|-----|
| | Ballista | Fires heavy bolts, no power, slow |
| | Gun turret | Standard turret, requires power, auto-targets |
| | Foam nozzle | Extinguishes fires in area, uses foam canisters |
| | Tesla coil | Chain lightning, hits multiple targets |
| | Mortar | Long range explosive, manual reload |
```

Traps -- Spot Check (5)

```
| Done | Trap | Test |
|-----|-----|-----|
| | Spike trap | Damages creature on walkover, rearms |
| | Bear trap | Immobilizes + damages |
| | Incendiary trap | Sets area on fire on trigger |
| | EMP mine | Electromagnetic burst on trigger |
| | Snare trap | Holds small creatures |
```

Fortifications (4)

```
| Done | Cover | Test |
|-----|-----|-----|
| | Sandbag | 40% cover, cheap |
| | Barricade | 60% cover |
| | Steel barrier | 75% cover |
| | Bunker | 85% cover |
```

14. RAIDS & THREATS

Raid System (5)

```
| Done | Feature | Test |
|-----|-----|-----|
| | Heat signature scaling | Reactor + colonists + cores -> raid budget |
| | Raid spawning | Enemies spawn from map edges |
| | Retreat mechanic | Raiders flee at 50% casualties (except swarms) |
| | Raid alert | RAID INCOMING alert triggers, clickable |
| | Post-raid effects | Hope/discontent shift, prisoner capture |
```

Raid Types (4)

```
| Done | Type | Test |
|-----|-----|-----|
| | Beast assault | Creature wave attacks colony |
| | Siege | Enemies build up before attacking |
| | Coordinated | Multi-direction organized attack |
| | Swarm (day 30+) | 3x budget, all 4 directions, no retreat |
```

Humanoid Raiders (3 factions)

```
| Done | Faction | Test |
|-----|-----|-----|
| | Outlaw raiders | Thug/brawler/gunner/marksman species spawn |
| | Scavenger raiders | Militia/scrapper/sharpshooter species spawn |
| | Mammona/MasTema | Enforcer/heavy/operative/sniper/breacher spawn |
```

Creature Threats (4)

```
| Done | Feature | Test |
|-----|-----|-----|
| | Natural spawning | Creatures appear at map edges scaled to difficulty/day |
| | Creature AI states | Idle->wander->flee->chase->attack transitions |
| | Lair spawning | Lairs periodically spawn creatures nearby |
| | Aggression range | Creatures aggro within defined radius, leash back |
```

15. DISEASE & MEDICAL

Diseases (5)

```
| Done | Disease | Test |
|-----|-----|-----|
| | Frostlung | Contracts from cold exposure, severity races immunity |
| | Blackrot | Severity vs immunity race, symptoms apply debuffs |
| | Ice Plague | Contagious, spreads to nearby colonists |
| | Tissue Creep | Contracts, progresses, treatable |
| | Spore Sickness | Contracts, progresses, treatable |
```

Medical Mechanics (5)

```
| Done | Feature | Test |
|-----|-----|-----|
| | Immunity race | Severity vs immunity both tick toward 100 |
| | Treatment | Medical colonist treats -> boosts immunity, slows severity |
| | Lethality | Lethal diseases kill if severity hits 100 first |
| | Symptom debuffs | Sick colonists have work speed/morale penalties |
| | Reinfection window | Recently recovered colonists have temporary immunity |
```

Surgery (5)

```
| Done | Feature | Test |
|-----|-----|-----|
| | Prosthetic install | Peg leg, wooden arm installable on surgery table |
```

		Bionic install		Bionic leg/arm/eye installable, stat bonuses	
		Organ harvest		Heart/lung/kidney/liver/eye harvestable (hope penalty)	
		Amputation		Damaged limbs removable	
		Surgery risk		Low medical skill -> failure chance -> additional wounds	

16. LOGISTICS & AUTOMATION

Conveyors (4)

	Done		Feature		Test	
	-----		-----		-----	
			Belt transport		Items move along belt in placed direction	
			Splitter		Alternates output between two belts	
			Belt freezing		Belts below -20C stop moving items	
			Belt + inserter chain		Items flow from belt -> inserter -> machine	

Inserters (3)

	Done		Type		Test	
	-----		-----		-----	
			Basic inserter		Moves items between belt and machine, slow	
			Fast inserter		2x speed of basic	
			Filter inserter		Only moves specified item types	

Pipe Network (6)

	Done		Feature		Test	
	-----		-----		-----	
			Fluid transport		Pipes carry water/oil/ichor/fuel/coolant/waste	
			Gas transport		Ducts carry oxygen/co2/steam/toxic_gas	
			Tank storage		Fluid/gas tanks store and equalize	
			Pipe freezing		Pipes freeze below threshold -> burst -> spill	
			Insulated pipes		Resist freezing	
			Dual union-find		Fluid and gas networks independent	

Pipe Processors (7)

	Done		Processor		Test	
	-----		-----		-----	
			Water pump		Extracts water from ice tiles	
			Oil refinery		Processes crude oil into fuel	
			Coolant refiner		Produces coolant from materials	
			Ichor converter		Processes eldritch ichor	
			Waste processor		Breaks down industrial waste	
			Gas separator		Separates gas mixtures	
			Steam boiler		Generates steam from water + heat	

17. RESEARCH & PROGRESSION

Research System (5)

	Done		Feature		Test	
	-----		-----		-----	
			Research bench		Colonist works at bench, progress advances	
			5-tier tree		All 58 nodes visible, tier prerequisites enforced	
			Node selection		Click node -> research begins, cancel button works	
			Unlock effects		Completed research unlocks buildings/recipes/features	
			Progress persistence		Research progress survives save/load	

Research Tiers -- Spot Check (5)

	Done		Tier		Sample Node		Test	
	-----		-----		-----		-----	
			Tier 1		basic_construction		Unlocks early buildings	
			Tier 2		agriculture		Unlocks farming buildings/crops	
			Tier 3		automation		Unlocks conveyors/inserters	
			Tier 4		missile_systems		Unlocks missile silo + warheads	
			Tier 5		shuttle_engineering		Unlocks launch_pad victory building	

18. TRADE & ECONOMY

Merchant System (4)

	Done		Feature		Test	
	-----		-----		-----	
			Merchant arrival		Storyteller triggers merchant visit	
			Merchant types		Scavenger/equipment/exotic with distinct inventories	
			Buy/sell		Player buys/sells via Trade panel, thermal cores as currency	
			Departure timer		Merchant leaves after timer expires	

Factions (5)

	Done		Faction		Test	
	-----		-----		-----	
			Mammona Logistics		Reputation tracks, affects trade prices	
			MasTema Ops		Reputation tracks, affects raid multiplier	
			Scavenger Crews		Reputation tracks	
			Ruin Delvers		Reputation tracks	
			Rim Runners		Reputation tracks	
			Reputation effects		Hostile (<-50) -> raids, Allied (>50) -> trade bonuses	

19. EXPLORATION & EXPEDITIONS

Expedition System (5)

Done	Feature	Test
	Launch expedition	Select colonists + destination, expedition starts
	Per-planet destinations	Each planet has its own destination catalog
	Party composition	1-3 colonists
	Success calculation	Skills + equipment + risk -> outcome tier
	Loot generation	Success returns resources/items to colony

Per-Planet Expedition Destinations (49)

Done	Planet	Destinations	Test
	Erebus	Frozen Wastes, Ice Caves, Predator Territory, Abandoned Outpost, Glacier Peak, Thermal Vents, Deep Rift, Precursor...	
	Rhea-2	Dune Sweep, Canyon Expedition, Oasis Search, Buried Ruins, Sun Temple, Deep Aquifer	6 destinations with desert r...
	Morvos	Acid Flats, Fungal Forest, Sealed Bunker, Spore Caves, Acid Lake Shore	5 destinations with acid-themed rewards
	Nerthus-9	Reef Dive, Island Hop, Shipwreck Dive, Deep Trench, Volcanic Vent	5 destinations with ocean rewards
	Paxtera Prime	Meadow Forage, Forest Expedition, Hill Quarry, Raider Camp, Old Settlement	5 destinations with temperate ...
	Nemaea	Surface Sweep, Wreckage Field, Automaton Graveyard, Dyson Fragment, Deep Vault	5 destinations with metal/compone...
	Gaia A^1x	Meadow Gathering, River Valley, Deep Woods, Corruption Edge, Heart Chamber	5 destinations with Baldrungen the...

Expedition View (4)

Done	Feature	Test
	Mini-tilemap	Expedition map visible in E panel
	Fog of war	Unrevealed areas dark, reveal on exploration
	POI markers	Entrance, loot, encounter, objective markers visible
	Progress timer	Countdown visible, expedition completes on finish

Vehicles (CUT FROM SCOPE)

Done	Feature	Test
	Vehicle construction (CUT)	Archived stub, not in shipped scope
	Expedition bonus (CUT)	Vehicle modifiers removed from expeditions
	Vehicle workbench (CUT)	Build flow removed from shipped scope

20. CREATURES & WILDLIFE

Creature Behavior (5)

Done	Feature	Test
	Spawning	Creatures appear at map edges, scaled by day/difficulty
	AI state machine	Idle->wander->flee->chase->attack works
	Pack behavior	Wolves/swarms move in groups
	Loot drops	Killed creatures drop meat/hide; special encounters can drop explicit salvage
	Corpse creation	Dead creatures leave corpses for butchering

Erebus Creatures (52 species)

Done	Tier	Species (sample)	Count	Test
	Small	Frost Hare, Ice Fox, Snow Grouse, Cinder Mite, Heat Skipper	5	Flee on sight, low HP
	Medium	Tundra Wolf, Glacier Bear, Ice Stalker, Ice Brute, Snow Ape, Stalker, Shade, Dire Wolf, Mammoth, Sabertooth, Char ...		
	Megafauna	Frost Titan, Thermal Wurm, Glacial Leviathan, Ancient Brute, Alpha Stalker, Mountain Titan, Ice Colossus, Storm...		
	Eldritch	The Hungering, The Pale Thing, That Which Sleeps, Fleshwalker, The Thermophage	5	Fear aura, night hunter, br...
	Eldritch Livestock	Gore Shoat, Weeping Calf, Husk Pup, Void Calf, Pit Wyrms, Bile Mold, Thorn Polyp, Nerve Cluster, Rot Bl...		
	Swarm	Frost Beetle, Ice Locust, Frost Wurm, Spawning, Skitterer, Permafrost Rat, Nerve Worm, Nerve Tick	8	Pack spawn...

Rhea-2 Desert Creatures (12 species)

Done	Tier	Species	Test
	Small	Sand Lizard, Dust Scarab, Canyon Sparrow	Flee, heat resistant
	Medium	Dune Stalker, Sand Viper, Ridge Raptor, Sand Wurm, Sun Scorpion, Wasteland Hyena	Aggressive, heat resistant, bre...
	Megafauna	Desert Colossus, Heat Drake, Dune Leviathan	Boss-tier, breacher, heat resistant

Morvos Acid Creatures (12 species)

Done	Tier	Species	Test
	Small	Acid Mite, Spore Crawler, Slime Beetle	Flee, acid immune
	Medium	Corrosion Hound, Acid Spitter, Fungal Stalker, Bile Brute, Caustic Wurm, Plague Carrier	Aggressive, acid immune,...
	Megafauna	Acid Titan, The Dissolvent, Mire Colossus	Boss-tier, acid immune, breacher

Nerthus-9 Ocean Creatures (12 species)

Done	Tier	Species	Test
	Small	Tide Crab, Reef Fish, Kelp Drifter	Flee, aquatic
	Medium	Depth Lurker, Reef Shark, Pressure Eel, Kraken Spawn, Abyssal Hunter, Barnacle Titan Juvenile	Aggressive, aquati...
	Megafauna	Storm Leviathan, The Depth Mother, Tidal Colossus	Boss-tier, aquatic, breacher

Paxtera Prime Temperate Creatures (12 species)

Done	Tier	Species	Test
	Small	Field Hare, Grain Bird, Orchard Snake	Flee, temperate fauna

| | Medium | Timber Wolf, Plains Bear, Wild Boar, Feral Bull, Apex Cat, Razorback Hog | Territorial, pack behavior, breacher (f...
| | Megafauna | Mammona Enforcer Mech, Great Elk, Territorial Megabear | Corporate mech + natural mega creatures |

Nemaea Automaton Creatures (11 species)

Done	Tier	Species	Test
	Small	Scout Drone, Maintenance Bot	Mechanical, vacuum+radiation immune, drop components/metal
	Medium	Patrol Automaton, Enforcer Unit, Salvage Drone, Siege Automaton, Hunter-Killer, Containment Unit	Mechanical, no ...
	Megafauna	Titan Automaton, The Warden, Dyson Sentinel	Boss-tier mechanical, breacher, fear aura (warden)

Gaia A^1x Creatures (12 species)

Done	Tier	Species	Test
	Small	Forest Rabbit, Songbird, Meadow Vole	Passive temperate fauna
	Medium (natural)	Timber Predator, Brook Bear, Grove Stalker	Pack hunter, territorial, night hunter
	Medium (Baldrungen)	Husk Crawler, Bone Beetle, Rot Wasp	Insectoid undead, no retreat, pack spawns, breacher (bone beetl...
	Megafauna	Brood Mother, The Emergence, Baldrungen Tendril	Massive insectoid horror spawns, breacher, fear aura

Xenolith Species (4)

Done	Species	Test
	Xenolith Queen	Eldritch-tier boss, 2000 HP, 50% armor, spawns 4 larvae on death, vacuum+radiation+acid immune
	Xenolith Drone	Megafauna, 300 HP, pack 2-5, vacuum+radiation immune
	Xenolith Larva	Swarm-tier, 40 HP, pack 4-10, matures into drone if not killed
	Xenolith Spore	Small passive, 15 HP, drifts and latches onto biomass

Lairs (3)

Done	Feature	Test
	Lair generation	Lairs placed during world gen
	Creature spawning	Lairs periodically spawn creatures nearby
	Lair destruction	Attacking lair destroys it, stops spawns

Taming (5)

Done	Feature	Test
	Tame attempt	Colonist attempts to tame wild creature
	Tame success	Skill check vs wildness -> creature becomes tamed
	Role assignment	Tamed animal assigned: livestock, hauler, guard
	Breeding	Tamed animals can breed (checkBreeding called)
	Resource yield	Livestock produce leather, meat

21. TILE TYPES

Base Tiles (17)

Done	Tile	Test
	Void	Impassable border
	Snow	Walkable, buildable
	Ice	Walkable, not buildable
	Rock	Solid, minable
	Permafrost	Walkable, buildable
	Dirt	Exposed from heat, walkable, buildable
	Wood/Stone/Metal Floor	Walkable, insulated per material
	Wood/Stone/Metal Wall	Solid, insulation per material
	Door	Walkable, partial gas permeability
	Water	Not walkable
	Lava Vent	Natural heat source
	Debris	Walkable, buildable
	Tree	Solid, minable for wood
	Ore Vein	Solid, minable for metal
	Insulated Wall/Floor/Door	High insulation, sealed door blocks gas+water

Underground Tiles (14)

Done	Tile	Test
	Deep Rock	Dense rock at edges, minable for stone
	Underground Rock	Excavatable cave stone
	Underground Floor	Walkable, buildable
	Shaft Entrance	Connects surface to underground
	Wood/Stone/Reinforced Column	Structural support with span 8/14/26
	Fungal Floor/Wall	Bioluminescent shallow cave tiles
	Membrane Floor/Wall	Precursor mid-cave tiles
	Organ Floor/Wall	Deep cave pulsing organic tiles
	Growth Creep	Spreading biological growth

Vertical Connections (5)

Done	Tile	Test
	Stair Down	Connects to layer below
	Stair Up	Connects to layer above
	Stairwell (Both)	Bidirectional
	Channel	Open pit, falling hazard

| | Ramp Up | Low move cost upward connection |

Natural Features (8)

| Done | Tile | Test |

|-----|-----|-----|

| | Frozen River | Fishable, becomes water during thaw |

| | Geyser | Periodic steam + heat, radius 6, 40 power |

| | Hot Spring | Warm pool, morale bonus |

| | Frozen Lake | Fishable surface |

| | Frozen Sea | Walkable, not buildable |

| | Cave Entrance | Surface cave mouth, connects to depth 1 |

| | Tundra Marsh | Walkable, movement penalty 1.5x |

| | Volcanic Rock/Floor/Ash | Warm volcanic tiles |

Lead Tiles (3)

| Done | Tile | Test |

|-----|-----|-----|

| | Lead Wall | Radiation-shielded wall |

| | Lead Door | Radiation-shielded passage |

| | Lead Ore | Minable for lead |

Desert Tiles -- Rhea-2 (6)

| Done | Tile | Test |

|-----|-----|-----|

| | Sand | Walkable, buildable, 1.3x move penalty |

| | Dune | Solid, excavatable for stone |

| | Sandstone | Solid, minable for stone |

| | Oasis | Water source, not walkable, fishable |

| | Cactus | Solid, choppable for food |

| | Cracked Earth | Walkable, buildable |

Ocean Tiles -- Nerthus-9 (5)

| Done | Tile | Test |

|-----|-----|-----|

| | Ocean | Deep water, not walkable |

| | Shallows | Walkable (slow), fishable, 1.8x move penalty |

| | Coral | Solid, minable for stone |

| | Beach | Walkable, buildable |

| | Seaweed | Walkable, harvestable, 1.4x move penalty |

Underwater Natural Resources (6)

| Done | Tile | Test |

|-----|-----|-----|

| | Coral Deposit | Minable for stone + food bonus |

| | Mineral Nodule | Minable for metal (5 yield) |

| | Thermal Mineral | Minable for components (3 yield) |

| | Kelp Forest | Walkable, minable for food, 1.6x move penalty |

| | Sunken Wreck | Minable for metal (6) + components (2) bonus |

| | Brine Pocket | Minable for fuel (3 yield) |

Floating/Underwater Construction (5)

| Done | Tile | Test |

|-----|-----|-----|

| | Pontoon | Floating wood platform, 1.2x move penalty |

| | Dock | Reinforced floating platform |

| | Sealed Floor | Pressure-rated, blocks water/gas |

| | Sealed Wall | Pressure-rated, high insulation |

| | Glass Dome | Transparent, morale bonus, blocks water/gas |

Temperate Tiles -- Paxtera/Gaia (5)

| Done | Tile | Test |

|-----|-----|-----|

| | Grass | Walkable, buildable |

| | Fertile Soil | Growth bonus 0.5, walkable, buildable |

| | Deciduous Tree | Solid, choppable for wood (4 yield) |

| | Bush | Walkable, harvestable, 1.2x move penalty |

| | Flower Field | Decorative, walkable, buildable |

Vacuum Tiles -- Nemaea (4)

| Done | Tile | Test |

|-----|-----|-----|

| | Regolith | Lunar dust, walkable, buildable |

| | Crater | Impact crater, walkable, not buildable |

| | Metal Debris | Minable for metal (4 yield) |

| | Hull Plate | Minable for components (2 yield) |

Space Tiles (7)

| Done | Tile | Test |

|-----|-----|-----|

| | Space Asteroid | Minable for metal, blocks movement |

| | Space Debris | Passable, damages hull |

| | Space Star | Impassable, lethal |

| | Space Corona | Passable, extreme heat damage |

| | Dyson Sphere Intact | Solid, impassable |

| | Dyson Sphere Broken | Passable debris |
| | Space Gravity Well | Near-planet gravity, slows ships |

22. BOSSES & MEGABEASTS

Named Bosses (4)

Done	Boss	Test
	The Bull (ice brute king)	Multi-phase fight, escalating abilities
	Frost Titan boss	Multi-phase, telegraphed attacks (1s warning)
	Thermal Wurm boss	Multi-phase, unique mechanics
	Glacial Leviathan boss	Multi-phase, drops unique loot

Boss Mechanics (4)

Done	Feature	Test
	Phase transitions	Boss enters new phase at HP thresholds
	Telegraphed abilities	1s warning before special attacks
	Unique drops	Boss-specific loot on kill (titan_heart, wurm_scale, etc.)
	Damage scaling	Each phase increases damage/speed multipliers

Procedural Megabeasts (3)

Done	Feature	Test
	Body form generation	10 forms (serpent, spider, titan, etc.)
	Material variation	10 material types affect stats
	Attack variety	10 attack VFX types

23. FIRE & HAZARDS

Fire System (6)

Done	Feature	Test
	Fire ignition	Incendiary traps, meltdowns, campfires can start fires
	Fire spread	15% base chance to adjacent flammable tiles
	Heat emission	Burning tiles emit 30C heat
	CO2 generation	Fires produce CO2 in enclosed rooms
	Entity damage	Entities on fire tiles take damage
	Weather suppression	Rain -70%, snow -50% spread chance

Firefighting (3)

Done	Feature	Test
	Auto-extinguish tasks	Fire auto-creates extinguish jobs for colonists
	Foam nozzle turret	Auto-targets fires in range, uses foam canisters
	Foam ordnance	Foam grenade/bomb/missile suppress fires + block ignition

Flooding (4)

Done	Feature	Test
	Water sources	Mined ice, burst pipes, breached water tiles
	Room flooding	Water level rises in enclosed rooms
	Drowning	Colonists take damage in deep water
	Pumps	Sump pumps remove water from rooms

Pollution (3)

Done	Feature	Test
	Pollution sources	Reactor, machines, incinerators produce pollution
	Pollution overlay (F3)	Pollution levels visible
	Scrubber	Reduces pollution in nearby tiles

24. EXPLORATION & WORLD

Map Secrets (5)

Done	Secret	Test
	Frozen colonist	Cryopod discoverable, thawable
	Thing mimic	Fake colonist, reveals as threat
	Precursor artifact	Discoverable, provides bonus
	Sealed cache	Discoverable, contains loot
	Dormant nest	Discoverable, may spawn creatures

BioCaves (3 tiers)

Done	Tier	Depth	Test
	Fungal	1-2	Fungal growth, skitterers, 5% discover chance
	Precursor	3-4	Precursor structures, spawnlings, 15% chance
	Organ	5+	Organic growth, fleshwalkers, 30% chance

Terraform (5)

Done	Operation	Test
	Smooth terrain	Rough -> smooth tile conversion

	Clear terrain	Remove debris/growth
	Excavate	Mine rock/permafrost/deep rock
	Dig shaft	Create entrance to underground layer
	Lay floor	Place floor tile on dirt/cleared ground

25. ELDRITCH & ANOMALY

Eldritch Nodes (5)

Done	Feature	Test
	5 growth stages	Larva->Whelp->Juvenile->Mature->Ancient
	8 eldritch types	Flesh, ichor, chitin, void, serpent, bile, thorn, nerve, rot
	Resource production	Mature+ nodes produce resources via output buffer
	Mutation risk	Scales with stage, anomaly_sensitive colonists reduce risk
	Inserter integration	Resources extractable by inserters

Anomaly System (4)

Done	Feature	Test
	Anomaly level (0-100)	Rises from drilling, exploring, node growth
	Event thresholds	Level 20/40/60/80 trigger escalating events
	Boss awakening	Level 80+ -> "That Which Sleeps" awakens
	Win condition	Defeating boss unlocks extraction victory

Skinwalker Events (4)

Done	Feature	Test
	Edge spawning	Skinwalker spawns at map edge
	Calling phase (30s)	Eerie calls emitted
	Luring phase (60s)	Individual colonist lured toward edge
	Attack phase	Stalker/alpha_stalker/fleshwalker attacks
	Anomaly_sensitive resistance	Sensitive colonists resist the lure

26. ENDGAME & VICTORY

Victory Paths (4)

Done	Path	Building	Test
	Mammona Claim	transmission_array	3-day charge -> final assault wave -> corporate claim ending
	Escape	launch_pad	Charge -> evacuate survivors -> escape ending
	Seal It	sealing_apparatus	Seal Erebus back down -> containment ending
	Mammona Extraction	extraction_beacon	Defeat boss first -> activate -> corporate extraction ending

Endgame Mechanics (3)

Done	Feature	Test
	Build phase	Endgame building constructible (expensive)
	Charge phase	3 game-day charging period
	Final challenge	Transmission array triggers final assault before victory

Game Over (3)

Done	Feature	Test
	All colonists dead	Game over triggers
	Stats display	Days survived, peak pop, cause of death, wealth
	Colony legacy	Dead colony saved as legacy record for future playthroughs

27. FARMING & FOOD

Agriculture (5)

Done	Feature	Test
	Farm plot placement	Farm plots placeable in heated rooms
	Crop selection	12 crops selectable via Farm panel
	Growth mechanics	tempFactor x lightFactor x waterFactor progression
	Harvest	Mature crops harvestable by colonists
	Sun lamp requirement	Indoor crops need sun lamp for light

Crops -- Spot Check (4)

Done	Crop	Test
	Ice rice	Fast grow (200s), low yield
	Frost potatoes	Medium grow (360s), reliable
	Tundra corn	Slow grow (600s), high yield
	Alien fungus	Grows in dark (no sun lamp needed)

Food Chain (4)

Done	Feature	Test
	Raw -> cooked	Kitchen converts raw meat/crops to meals
	Spoilage	Food items rot over time, faster in heat
	Smokehouse	Meat -> jerky (long shelf life)
	Feast recipe	High-quality meal from multiple ingredients

28. COLONIST GROWTH & RECRUITMENT

Colony Growth Sources (4)

Done	Source	Test
	Prisoner capture	25% chance from raid kills -> prisoner -> recruit
	Refugee events	Storyteller triggers refugee arrival
	Cloning vat	300s grow time, 30 food + 2 components -> new colonist
	Radio beacon	Passive wanderer attractor (also increases heat sig)

Slavery (CUT FROM SCOPE)

Done	Feature	Test
	Prisoner -> slave (CUT)	Archived stub, not a shipped system
	Slave work (CUT)	Archived stub, not a shipped system
	Escape risk (CUT)	Archived stub, not a shipped system
	Morale penalty (CUT)	Legacy penalty entry only, not reachable in normal play
	Free/sell/process (CUT)	Archived stub, not a shipped system

Strange Moods (4)

Done	Feature	Test
	Mood trigger	3% per colonist per day -> colonist claims workshop
	Material demands	Colonist demands specific materials
	Artifact creation	30s crafting -> legendary item with bonuses
	Failure	Timeout (120s) -> mental break or death

29. SAVE/LOAD & PERSISTENCE

Save System (5)

Done	Feature	Test
	Quick save (F5)	All game state written to disk, toast confirms
	Quick load (F9)	Game state restored from disk, toast confirms
	Auto-save	Every 5 real minutes, no player action required
	ECS serialization	All entities + components preserved
	Tilemap serialization	Tile data + temps + room IDs per depth layer

Save/Load Integrity (10)

Done	System	Test
	Colonist state	Needs, skills, traits, wounds, disease survive save/load
	Building state	All buildings, recipes, durability survive
	Thermal state	Room temperatures, reactor state survive
	Power grid	Generator fuel, battery charge, grid connectivity survive
	Research progress	Current research + completed nodes survive
	Pipe network	Fluid/gas levels, tank contents, processor state survive
	Active raid	Raid in progress survives save/load
	Weather/season	Current weather, season, day/time survive
	Planet state	GameState.planet, landing zone, planet-specific systems (pressure/corrosion/baldrungen) survive
	Easter egg state	Blood ocean active, blood rain fired flags survive

30. EASTER EGGS

Fischbach / Markiplier Easter Egg (7)

Done	Feature	Test
	Fischbach detection	System detects any living colonist with last name matching /[Ff]ischbach\$/
	Blood rain trigger (Erebus)	First rain event becomes blood_rain if Fischbach exists (weather.lua)
	Blood ocean (Nerthus-9)	Day 2 check: ocean tiles turn blood red, water rises
	Blood rain (other planets)	Day 2 check: 180s blood rain forced on non-ocean planets
	Ocean rising	Blood ocean active: water injected into surface tiles every 2s
	Submersible curse	Fischbach in submersible at depth causes durability drain + blood water leak
	Ocean calms	Fischbach dying at depth > 0 deactivates blood ocean; surface death = permanent

Blood Ocean Color Swaps (4)

| Done | Tile | Blood Color | Test |
|-----|-----|-----|
| | Ocean | (0.45, 0.05, 0.05) | Deep red |
| | Shallows | (0.55, 0.1, 0.1) | Shallow red |
| | Water | (0.5, 0.08, 0.08) | Blood water |
| | Seaweed | (0.35, 0.1, 0.05) | Dark red-brown |

31. MULTIPLAYER

Network (4)

Done	Feature	Test
	Host game	lua-enet P2P host initializes
	Join game	Client connects to host
	Shared control	Actions sync between players
	Inter-colony trade	Trade routes between player colonies

Multiplayer UI (2)

Done	Feature	Test
	Host/Join screen	**MISSING** -- No UI to create or join game
	Player list	**MISSING** -- No connected player display

32. MISC SYSTEMS

Deterioration & Maintenance (3)

Done	Feature	Test
	Building decay	Buildings lose durability over time
	Repair task	Colonists auto-repair damaged buildings
	Destruction	Building at 0 durability -> destroyed

Decorations & Room Quality (3)

Done	Feature	Test
	Decoration placement	Decorative items placeable in rooms
	Room impressiveness	Formula based on size + furnishing + cleanliness
	Morale bonus	Better rooms -> higher colonist morale

Filth & Cleaning (2)

Done	Feature	Test
	Filth accumulation	Dirty areas accumulate filth over time
	Clean job	Colonists clean filthy areas when assigned

Equipment (3)

Done	Feature	Test
	Weapon equip	Colonists can be assigned melee/ranged weapons
	Armor equip	Colonists can be assigned armor
	Thermal suits	Thermal suit/exosuit for expedition protection

Quest System (3)

Done	Feature	Test
	Quest generation	Procedural quests created
	Quest objectives	Objectives trackable and completable
	Quest panel	Panel exists, Q hotkey wired in main.lua

Elastic Difficulty (2)

Done	Feature	Test
	Difficulty scaling	Threat scales with colony wealth/population
	Adaptation	System adapts to player performance

KNOWN GAPS SUMMARY

#	Gap	Impact	Priority
1	~~Taming has no work AI job~~	**RESOLVED** -- Taming job exists in jobs.lua	~~HIGH~~
2	Multiplayer has no host/join UI	Full P2P backend unreachable by players	HIGH
3	Victory condition framing	Names in docs/UI can drift if not kept in sync with the 4 live endgame outcomes	LOW
4	~~Quest panel hotkey~~	**RESOLVED** -- Q key wired in main.lua and quest_panel.lua	~~LOW~~
5	Vehicle construction UI (CUT)	Archived scope item, not a shipped feature gap	LOW
6	Turret targeting priority	No UI to set turret priorities/ranges	LOW
7	~~Sprite art~~	**RESOLVED** -- 379 PNGs in assets/sprites/, lazy-loaded via sprites.lua	~~HIGH~~
8	Merchant caravan visuals	Minimal entity representation on map	LOW
9	Taming system backend-only	No work AI job for animal husbandry	MEDIUM
10	4 victory paths undiscoverable	No in-game guidance for players to find victory conditions	MEDIUM

TOTALS

Category	Items
Core game loop	17
Player controls & input	34
UI panels	56
Planet system	109
Per-planet mechanics	44
Colonist systems	63
Social & morale	33
Building & construction	72
Underwater & ocean buildings	22
Weather system (all planets)	35
Thermal & environment	23
Power system (incl. 38 generators)	44
Combat & defense	27
Raids & threats	16
Disease & medical	15
Logistics & automation	20

Research & progression	10
Trade & economy	10
Exploration & expeditions (49 destinations)	61
Creatures & wildlife (123 species)	105
Tile types (84 tiles)	85
Bosses & megabeasts	11
Fire & hazards	16
World exploration	13
Eldritch & anomaly	13
Endgame & victory	10
Farming & food	13
Colony growth	13
Save/load	15
Easter eggs	11
Multiplayer	6
Misc systems	16
GRAND TOTAL	**~1,058 testable items**
*Generated from codebase audit on 2026-03-16. Covers 7 planets, 123 creature species, 84 tile types, 38 power generators, 49 ex...